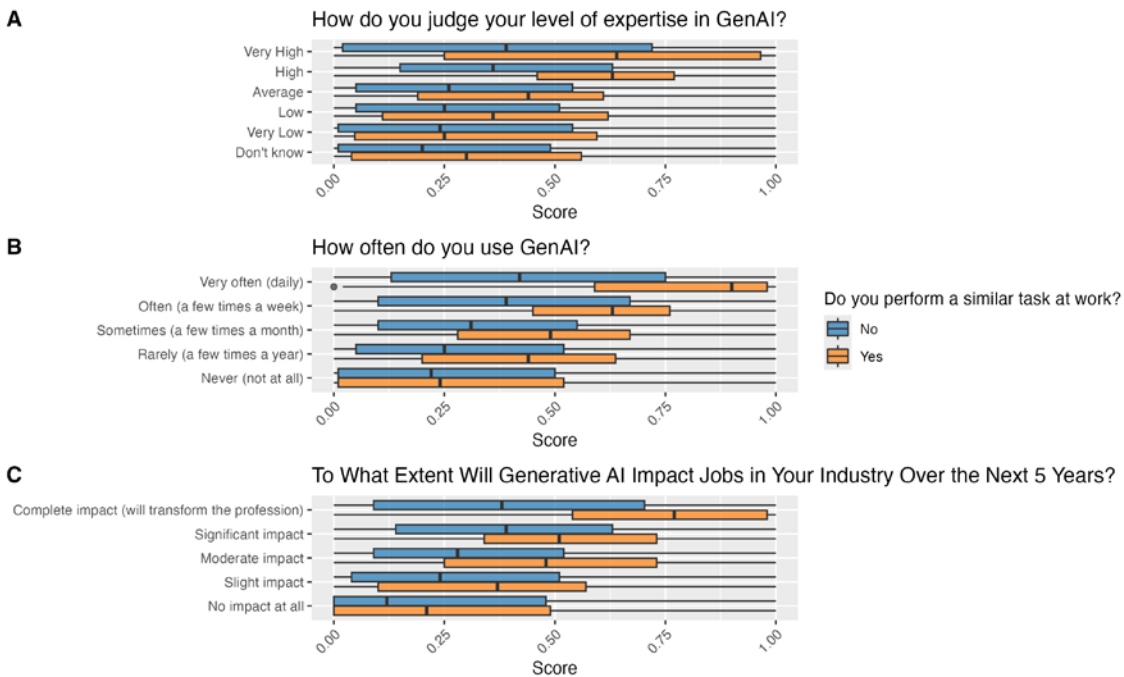


► **Figure 10. Scoring as a function of familiarity with GenAI and the scored task – distribution of individual scores**



One way to address this variation in scoring would be to apply corrective weights to responses, giving greater influence to respondent profiles likely to have more relevant expertise. For example, one could assign more weight to respondents reporting greater familiarity with GenAI or frequent usage or prioritize scores from individuals who perform similar tasks in their daily work. However, in our sample, this would introduce significant limitations. For 91.5% of the tasks, respondents did not report performing a similar task in their work – even when the task belonged to the same 1-digit occupational group as their current job. This is not surprising. For instance, within the “Professionals” group alone, the representative sample consisted of 368 tasks drawn from a pool of 8,715 tasks (Table 2), making exact task matching unlikely.

Moreover, adjusting for beliefs or usage would require strong theoretical assumptions about what constitutes “credible” scoring. For example, a respondent who believes their profession will be completely transformed by GenAI in the next five years may be highly informed, or may be expressing anxiety, speculation, or dystopian fears. Without richer data to distinguish between these interpretations, applying corrective weights risks introducing new biases.

Given the relatively small size of these specific subgroups in our dataset, we refrain from implementing such adjustments in the current analysis to avoid overfitting or compromising credibility. Instead, we flag this as an important area for future survey design, particularly in capturing respondent expertise and task familiarity more systematically.

For the present study, we treat the survey as a starting point, designed to capture a broad spectrum of perspectives reflecting varying degrees of familiarity with both tasks and GenAI. In many cases, a respondent’s understanding of the task itself may be more influential than their understanding of the technology, especially for concrete physical tasks (e.g. trimming flowers, assembling mechanical parts). Accordingly, we rely on mean scores across all respondents to reflect the collective judgment of a diverse sample grounded in workplace experience. To enhance

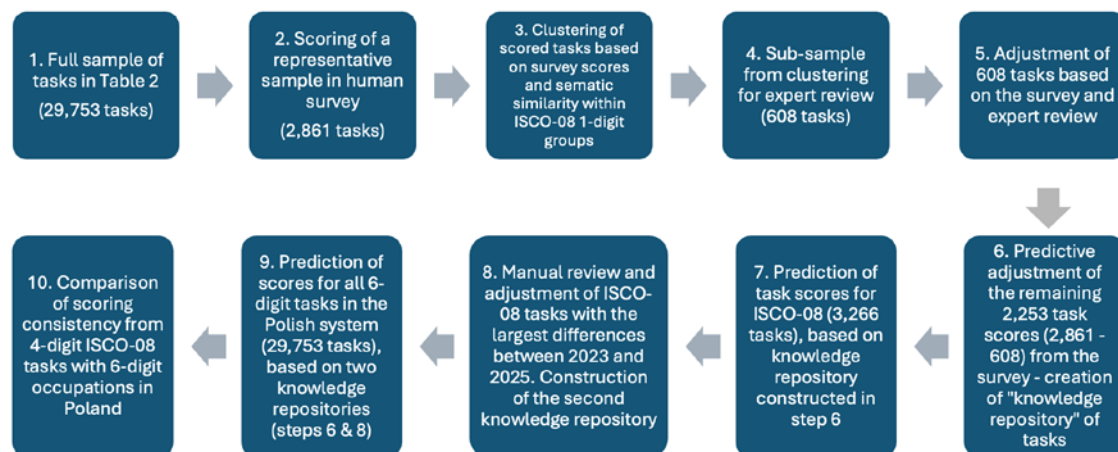
robustness, we supplement this with expert evaluation of a subset of tasks (section 5), ensuring that final scores reflect both practical insights and a realistic appraisal of current GenAI capabilities.

## ► 4 Expert validation survey

The expert group includes international specialists at the ILO and its partner organizations, NASK and the Polish Ministry of Family, Labour and Social Policy.<sup>9</sup> The primary criterion for expert selection is their understanding of labour markets, including the complexity of occupations and tasks, and their real-life applications. While the experts are expected to have familiarity with the current family of generative AI technologies and their applications, expertise in AI design or development is not a requirement. This approach ensures that the focus remains on the feasibility and practical implications of automation within existing technological constraints, as technology experts tend to overstate technological capacities and risks when evaluating broader applications, particularly in workplace settings (Karger et al., 2023). This tendency is evident in earlier automation studies, which often assigned high displacement potential to routine tasks or entire occupations, such as garment production. As de Mattos et al. (2020) illustrate, these technological predictions frequently failed to account for the challenges of automating tasks requiring high dexterity or handling complex materials, such as stitching pliable fabrics. Accordingly, the assessment of automation potential in our study prioritizes expertise of task execution and workplace realities over purely technological expertise.

Since the experts cannot review all 2,861 tasks used in the survey, we focus on a representative sub-sample, selected in such a way to subsequently permit adjustments to entire task clusters. The overall scheme of this score adjustment procedure is presented in Figure 11.

► Figure 11. Schema of the scoring stages



First, we cluster the tasks into four fixed intervals, based on the distribution of the mean scores from the survey.<sup>10</sup> We then use hierarchical clustering to group similar tasks within those clusters based on their semantic content. First, we assign embeddings to each task description using the *text-embedding-ada-002* model from OpenAI. Next, for each subgroup – defined by the combination of an occupational group at ISCO-08 1-digit level and a particular survey score cluster – we

<sup>9</sup> A full list of contributors is provided in the acknowledgements section.

<sup>10</sup>  $\mu \in [0, 0.1] \cup (0.1, 0.3] \cup (0.3, 0.5] \cup (0.5, 1]$